



山东大学

SDU OS Slide

A Typst slide template for SDU operating-system lectures

Guide and API reference

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Guide

SDU OS Slide recreates the visual language of the Shandong University operating-system course slides while keeping the source small and ordinary Typst. The public surface is intentionally narrow: import the setup function, write headings and content, and use a few overlay markers when a slide should build step by step.

Quick Start

```
#import "slide.typ": setup, pause, meanwhile, jump, theme

#show: setup.with(
  title: "SDU OS Slide",
  subtitle: "Slide for SDU OS",
  author: "arshtyi",
  term: "2026 Spring",
  date: datetime.today(),
  handout: false,
  sidebar-ring-style: 1,
)

= Chapter
== Slide Title

Always visible.

#pause

Visible on the next overlay.
```

Import path

Use a relative import when working from this repository.

Document Model

- `= Section` starts a new slide and appears as a top-level sidebar entry.
- `== Slide` starts a new slide, appears in the sidebar, and is highlighted while active.
- `=== Topic` starts a new slide but is rendered as a centered in-slide title instead of a sidebar entry.
- Deeper headings also inherit the slide-break rule, but they are best kept rare.

The right sidebar is built from outlined level 1 and level 2 headings. Each entry is a link to the original slide location.

Visual Frame

`setup` renders a cover page, page frame, SDU logo, right sidebar, and decorative lower-right ring. The option `sidebar-ring-style` controls the ring/sidebar layering:

Value	Effect
1	Current style. The ring is drawn above the sidebar table. This is the default.
2	Final style. Normal row fills sit below the ring, active row fill sits above it, and gutters mask the ring.

Theme constants are exported as `theme`, so user documents can read values such as `theme.colors.primary` or `theme.page.sidebar-width` without duplicating them.

Overlays

`pause`, `meanwhile`, and `jump` are small content markers processed by the theme's overlay renderer.

Marker	Meaning
<code>#pause</code>	Advance by one overlay step. Later content is hidden until the next step.
<code>#meanwhile</code>	Reset following content back to the first overlay step. Useful for parallel build sequences.
<code>#jump(n)</code>	Low-level overlay control. Use <code>relative: true</code> to move relative to the current step.

Set `handout: true` to output only the final overlay state of each slide.

Build

The template is plain Typst. Useful smoke tests are:

```
typst compile --root . template/template.typ template/typst-sdu-os-slide.pdf
typst compile --root . docs/manual.typ docs/manual.pdf
```

Reference

slide

Public entry point. Most users import from this file only.

theme

Main setup function and page-level theme wiring.

- `setup()`

setup

Applies the SDU OS slide theme to the document.

The setup function renders a cover page, configures the slide canvas, installs the heading styles, and enables overlay rendering through `pause`, `meanwhile`, and `jump`.

Headings always start a new page. Level 3 and deeper headings are rendered as centered in-slide titles.

- `author (str)`: Author name shown on the cover and sidebar.
- `term (str)`: Course term shown on the cover.
- `title (str)`: Presentation title shown on the cover.
- `subtitle (str)`: Subtitle shown on the cover and sidebar.
- `date (datetime)`: Date shown in the sidebar.
- `handout (bool)`: Whether to render only the final overlay state of each slide.
- `sidebar-ring-style (int)`: `1` keeps the old ring layering; `2` interleaves the ring with sidebar fills, active row, and gutters.
- `body (content)`: Presentation body.

Parameters

```
setup(  
    author,  
    term,  
    title,  
    subtitle,  
    date,  
    handout,  
    sidebar-ring-style,  
    body  
) -> content
```

overlays

Overlay markers and the renderer used by `setup`.

- `jump()`
- `render-overlays()`

Variables

- `pause`
- `meanwhile`

jump

Creates an overlay jump marker.

`jump` is the common primitive behind `pause` and `meanwhile`. With `relative: true`, `n` moves the current overlay step by a relative amount. With `relative: false`, `n` sets the current overlay step absolutely.

- `n` (int): Relative or absolute overlay step.
- `relative` (bool): Whether `n` is a relative offset. Defaults to `false`.

Parameters

```
jump(
  n,
  relative
) -> content
```

render-overlays

Renders a document body with slide overlays.

The renderer splits the body at headings and at explicit `pagebreak()` calls. In presentation mode, each resulting slide is duplicated once for each overlay step introduced by `pause`, `meanwhile`, or `jump`. In handout mode, only the final overlay state of each slide is rendered.

- `handout` (bool): Whether to render only final overlay states. Defaults to `false`.
- `body` (content): Document body to render.

Parameters

```
render-overlays(
  body,
  handout
) -> content
```

pause content

Advances to the next overlay step.

Content after `pause` is hidden on the current overlay and appears on the next overlay. It is equivalent to `jump(1, relative: true)`.

meanwhile content

Resets subsequent content to the first overlay step.

Content after `meanwhile` appears together with the beginning of the slide, while earlier paused content keeps its own reveal timing. It is equivalent to `jump(1)`.

foundation-theme

Shared fonts, colors, and page geometry.

Variables

- `theme`

theme dictionary

Shared visual constants for the SDU OS slide theme.

The dictionary contains font families, colors, and page dimensions used by the cover, sidebar, frame, and body text.

cover

Cover-page component.

- `render-cover()`

render-cover

Renders the opening cover page.

The cover uses the SDU and SDU CS logos, decorative circles, and the theme's cover font and color tokens. It also resets the page counter so the first content slide starts from page 1.

- `title (str)`: Main title shown in the center of the cover.
- `subtitle (str)`: Subtitle shown below the title.
- `author (str)`: Author name shown in the lower badge.
- `term (str)`: Course term shown in the lower badge.

Parameters

```
render-cover(  
  title,  
  subtitle,  
  author,  
  term  
) -> content
```

sidebar

Right-side outline, active-row detection, and sidebar/ring layering helpers.

- `format-outline-heading()`
- `find-active-subheading()`
- `build-outline-cells()`
- `render-sidebar()`
- `render-sidebar-gap-mask()`

format-outline-heading

Formats a level 1 or level 2 heading for the sidebar outline.

The generated heading text is linked to the original heading location. Level 1 headings use bold body text; level 2 headings keep regular body text after the numbering.

- `heading-node (content)`: Heading node returned by `query(heading)`.

Parameters

```
format-outline-heading(heading-node) -> content
```

find-active-subheading

Finds the active level 2 heading for a page.

A subheading is active from the page where it appears until the next outline heading starts. This keeps overlay pages inside the same highlighted sidebar row.

- `outline-headings (array)`: Queried level 1 and level 2 headings.
- `current-page (int)`: Current page number.

Parameters

```
find-active-subheading(
  outline-headings,
  current-page
) -> none content
```

build-outline-cells

Builds table cells for the sidebar outline.

The returned dictionary contains the cells and the table row that should be filled with the active color.

- outline-headings (array): Queried level 1 and level 2 headings.
- active-subheading (none, content): Active level 2 heading, if any.
- show-text (bool): Whether to render the row labels.

Parameters

```
build-outline-cells(
  outline-headings,
  active-subheading,
  show-text
) -> dictionary
```

render-sidebar

Renders a sidebar table layer.

The `fill-mode` controls which row backgrounds are drawn: `"full"` draws the normal sidebar fills, `"base"` skips the active row, and `"active"` only draws the active row fill. This lets the theme interleave the decorative ring between normal row fills and the active row.

- title (str): Sidebar title, usually the presentation subtitle.
- author (str): Author name shown in the footer.
- date (datetime): Date shown in the footer.
- fill-mode (str): Either `"full"`, `"base"`, or `"active"`.
- show-text (bool): Whether to render title, outline labels, and footer.

Parameters

```
render-sidebar(
  title,
  author,
  date,
  fill-mode,
  show-text
) -> content
```

render-sidebar-gap-mask

Masks sidebar gutters above the decorative ring.

The mask follows the same row-gutter sequence as the sidebar table and adds a thin right and bottom edge so the ring remains clipped by the table gaps.

- width (length): Width of the sidebar block.
- height (length): Height of the sidebar block.
- edge-gutter (length): Thickness of the right and bottom masks.

Parameters

```
render-sidebar-gap-mask(
  width,
  height,
  edge-gutter
) -> content
```

draw

Low-level drawing helpers.

- `place-circle()`

place-circle

Places a CeTZ circle by its center point.

Typst's `place` works from a top-left anchor in this template, while the visual design is easier to describe with circle centers. This helper converts center coordinates and radius to the correct placed CeTZ canvas.

- center-x (length): Horizontal center position.
- center-y (length): Vertical center position.
- radius (length): Circle radius.
- style (arguments): Additional CeTZ circle styling such as `fill` or `stroke`.

Parameters

```
place-circle(
  center-x,
  center-y,
  radius,
  ..style
) -> content
```